

Homework 6

Connor Mooneyhan

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1. Show that every norm on $\mathbb{R}$ is of the form $\|x\| = c \cdot |x|$ for some real number $c > 0$; here $|x|$ denotes the absolute value of x .

Proof. Let $x, y \in \mathbb{R}$, and let $\|\cdot\|$ be a norm on $\mathbb{R}$. Then

$$\begin{aligned} |x| \cdot \frac{\|y\|}{|y|} &= \left| \frac{x}{y} \right| \|y\| \\ &= \left\| \frac{x}{y} \cdot y \right\| \\ &= \|x\|, \end{aligned}$$

where the second equality comes from homogeneity. Rewriting the equality above, we see that for *any* $x, y \in \mathbb{R}$, there is a shared common ratio

$$\frac{\|x\|}{|x|} = \frac{\|y\|}{|y|}.$$

Call that common ratio c , and note that $c > 0$ because of the positivity of $\|\cdot\|$ and the absolute value function. Then we have $\|x\|/|x| = c$, or in other words, $\|x\| = c \cdot |x|$ for all $x \in \mathbb{R}$ as desired. $\square$

2. Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$. Note that $\|\cdot\|$ is, by definition, a function from $\mathbb{R}^n$ to $\mathbb{R}$. Show that this function is continuous with respect to the box metric of $\mathbb{R}^n$.

Hint: You should first convince yourself that it suffices to show $\|x_n\| \rightarrow 0$ whenever $x_n \rightarrow 0$ with respect to the Euclidean metric.

Proof. Indeed, we will show that if $x_m \rightarrow 0$ w.r.t. the Euclidean metric, $\|x_m\| \rightarrow 0$. Given some such (x_m) , write $x_m = \sum_{i=1}^n (x_m)_i e_i$ for each x_m where the e_i are the standard basis vectors in $\mathbb{R}^n$. Since $x_m \rightarrow 0$, each $(x_m)_i \rightarrow 0$ since $(x_m)_i \leq \|x_m\|_\infty$ for all i . Then

$$0 \leq \|x_m\| \leq \sum_{i=1}^n |(x_m)_i| \cdot \|e_i\| \rightarrow 0,$$

so $\|x_m\| \rightarrow 0$. $\square$

3. Let S^n be the unit sphere with respect to the box metric. More precisely,

$$S^n = \{x \in \mathbb{R}^n : \|x\|_\infty = 1\}$$

where $\|\cdot\|_\infty$ denotes the box norm. Use the conclusion of Question 2 to deduce that $\|\cdot\|$ achieves a minimum value and maximum value on S^n .

Proof. (I'm going with your notation even though we would usually situate S^n within $\mathbb{R}^{n+1}$.) Since $\|\cdot\|_\infty$ on $\mathbb{R}^n$ is equivalent to the standard norm on $\mathbb{R}^n$, we can use Heine-Borel to see that S^n is a compact set since it is closed and it is bounded inside a standard ball of radius 5. Thus restricting any norm $\|\cdot\|$ to the domain S^n yields a continuous function whose domain is compact, which thus achieves a minimum and maximum value by the extreme value theorem. $\square$

4. Use the conclusion of Question 3 to show that any norm $\|\cdot\|$ on $\mathbb{R}^n$ is equivalent to the box norm $\|\cdot\|_\infty$: there exists $C > 1$ such that for all $x \in \mathbb{R}^n$ we have

$$\frac{1}{C}\|x\|_\infty \leq \|x\| \leq C\|x\|_\infty.$$

Proof. First write $x \in \mathbb{R}^n$ as $x = \sum_{i=1}^n x_i e_i$ where the e_i are the standard basis vectors. Then

$$\|x\| \leq \sum_{i=1}^n \|x_i\| \cdot \|e_i\| \leq \left(\sum_{i=1}^n \|e_i\| \right) \cdot \|x\|_\infty.$$

Thus if we let $c_1 = \sum_{i=1}^n \|e_i\|$, then $\|x\| \leq c_1\|x\|_\infty$.

On the other hand, let $c_2 = \min_{x \in S^n} \|x\|$ by (3). Since each $\|x\|$ in that set is nonzero, $c_2 > 0$. Now since $x/\|x\|_\infty \in S^n$ for any nonzero x , we have

$$\left\| \frac{x}{\|x\|_\infty} \right\| \geq c_2,$$

which implies that

$$c_2\|x\|_\infty \leq \|x\|.$$

Thus if we take $C = \max(1/c_2, c_1, 2)$, we get $C > 1$ by construction and

$$\frac{1}{C}\|x\|_\infty \leq c_2\|x\|_\infty \leq \|x\| \leq c_1\|x\|_\infty \leq C\|x\|_\infty.$$

$\square$

5. Read Section 1.2-3 of the book on l^p spaces, and use Equation (6) on Page 13 to prove the (integral) Holder's inequality: for all $f, g \in C[0, 1]$ and all $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$, it holds that

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

Proof. First, suppose that $\|f\|_p = \|g\|_q = 1$. Then

$$|f(x)g(x)| \leq \frac{|f(x)|^p}{p} + \frac{|g(x)|^q}{q}$$

by (6). Integrating this on both sides, we get

$$\|fg\|_1 \leq 1 = \|f\|_p \|g\|_q$$

as desired.

Now define

$$\tilde{f}(x) = \frac{|f(x)|}{\|f\|_p}, \quad \tilde{g}(x) = \frac{|g(x)|}{\|g\|_q} \in C[0, 1].$$

Then $\|\tilde{f}\|_p = 1$ and $\|\tilde{g}\|_q = 1$, so the first part shows our result. □